

Software Requirements Specification (SRS)

Project: ERP Management System (MERN Stack)

1. Purpose

Develop a secure, scalable ERP web application for managing products, customers, suppliers, sales orders, purchase orders, GRNs, invoices, and users with role-based access.

2. Scope

The system supports Admin, Sales, Purchase, and Inventory users. It provides JWT authentication, CRUD operations, dashboards, reports, PDF invoices, search, pagination, and responsive UI.

3. Users & Roles

Admin: full access. Sales: customers, sales orders, invoices. Purchase: suppliers, purchase orders, GRNs. Inventory: products and stock.

4. Functional Requirements

- Authentication: Register, Login, JWT, Logout.
- User Management: CRUD, role assignment.
- Product Management: CRUD, search, pagination.
- Customer & Supplier Management: CRUD.
- Sales Orders: create/update/view.
- Purchase Orders: create/update/view.
- GRN: receive goods linked to purchase orders and update stock.
- Invoices: generate, download PDF, view history.
- Dashboard: KPIs, charts, low stock alerts.

5. Non-Functional Requirements

Responsive UI, secure password hashing (bcrypt), JWT authorization, modular architecture, reusable components, REST APIs, validation, centralized error handling, performance via pagination.

6. Technology Stack

Frontend: React, MUI, Redux Toolkit. Backend: Node.js, Express.js. Database: MongoDB. Auth: JWT + bcrypt.

7. Database Collections

users, products, customers, suppliers, salesOrders, purchaseOrders, grns, invoices.

8. API Modules

Users, Products, Customers, Suppliers, Sales Orders, Purchase Orders, GRNs, Invoices.

9. Testing

Backend: Jest. Frontend: React Testing Library.

10. Deployment

Frontend on Vercel, Backend on Render, Database on MongoDB Atlas.